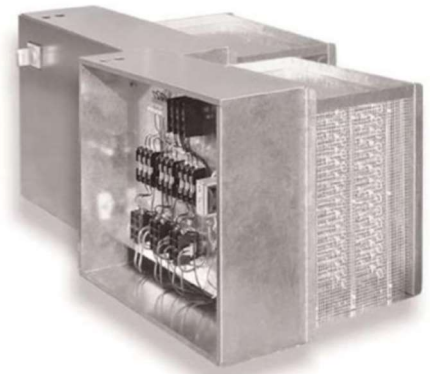


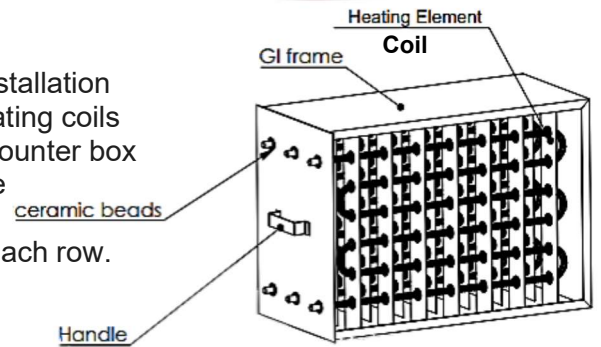
Electric Duct Heater

Duct heater is used to reheat air before it reaches the conditioned space to ensure that it reaches the desired temperature. It employs a set of heating elements (high resistance coil) electrically powered arranged in horizontal rows within its frame. These elements transform electrical energy into thermal energy via conduction and convection, resulting in a temperature rise.



Components:

The heater consists of a control box connected to a frame for installation on the air duct, with the control box outside the duct and the heating coils extending from the control box so that it is in the air path. The Counter box consists of a set of heating coils mounted inside the frame in the form of horizontal rows supported by disks of ceramic material that is resistant to heat. The disks are evenly distributed along each row.

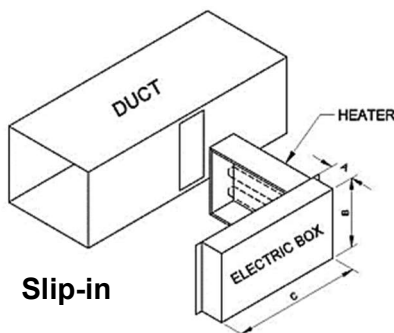


6-Rows, 5-Columns REH

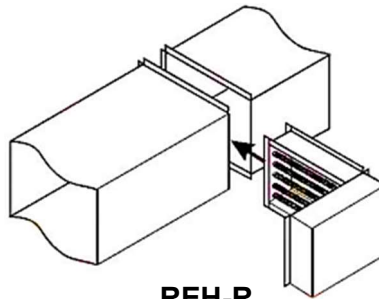
Frame:

The frame surrounds the heating coil and is made of galvanized steel with a thickness of 1 mm. It is compatible with the shape of the air duct. There are several types of heaters depending on the installation method with the duct, and each type has a different frame shape, as follows:

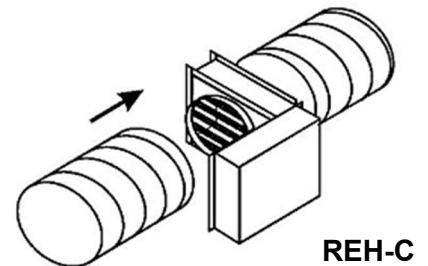
- **Slip-in heater:** A side opening is made in the duct and the frame and heating coil are inserted from it by pushing them inside. This is the most common type because it is easy to install.
- **Flange heater (REH-R):** It is installed on the duct using a flange 1 inch wide on both sides.
- **Round heater (REH-C):** Hat shaped frame with a round spigot attachment, suitable for installation on circular ducts.



Slip-in



REH-R



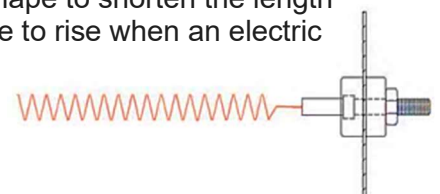
REH-C

- The customer only specifies the duct size (WxH) & heater capacity (Kw), and the heater is manufactured with dimensions that are compatible with this size, capacity & the selected model.

Heating coil:

The heating coil is a wire made of nickel-chrome alloy of high electrical resistance, which causes its temperature to rise when an electric current passes through it. There are 3 models of heating coils:

- **Open Coil:** A wire made of nickel-chrome alloy wound in a spiral shape to shorten the length and give it a high electrical resistance, which causes its temperature to rise when an electric current passes through it. **(Standard)**



- **Tubular Element:** A tubular element made of Nickel chrome alloy of 3/8" diameter containing a heating coil in magnesium oxide powder. Comes with 2 shapes U or W determined by the heater dimensions. **(Optional)**
- **Finned-Tubular Element:** Like the tubular element, the finned-tubular version features aluminum fins. These fins boost the surface area, enabling more efficient heat dissipation via convection. Two shapes, U and W, determined by the heater dimensions. **(Optional)**



Tubular Element



Finned-Tubular Element

Control box:

The control box is made of galvanized steel and contains the electrical components necessary to control the circuit.

Circuit components:

- **Circuit breaker:** The incoming wires to the heater and coming from the power source are connected directly to this breaker, which cuts off the current from the circuit in the event of an electrical short circuit or a sudden high load. The circuit breaker is the same type used in homes.
- **Transformer:** Converts the input electrical voltage of 220 or 380 to 24 volts, which is the required value for the control circuit.
- **Thermal switch:** Disconnects the circuit when the temperature rises to an unacceptable level. Two types of this switch are installed in the circuit: a type that is automatically reset when the temperature returns to the acceptable value and a type that is reset manually when the temperature reaches the critical value.
- **The automatic type is set to a temperature of 50° and the manual type to a temperature of 80°.**
- **The purpose of the manual switch is to require those involved to inspect the heater before restarting it.**
- **Flow switch:** Its function is to ensure that the circuit only operates when there is air flow inside the duct.
- **Contactors:** It takes the signal from the previous components to turn on and off the circuit.
- **Thermostat:** It is mounted inside the conditioned room to measure the temperature inside it. The user can set it to the desired temperature so that it compares between the measured temperature and the desired temperature and gives an indication to the heater to turn on or off. It is recommended to mount it inside the room at a height of 1.5 meters from the ground.
- **Disconnect switch:** A manual switch for turning on or off is mounted on the control box from the outside. It is used for manual testing or for disconnection in case of maintenance to increase safety.
- **Pilot light:** This light acts as an indicator of the heater's status whether it is in the on or off state.

The heater can be connected to the Building Management System (BMS) so that it takes the signal directly from the system. In this case, the thermostat can be dispensed with or used only as an indicator of the temperature without controlling it from it.

The disconnect switch and pilot light are accessories to the circuit, so they are added upon request.

How To order:

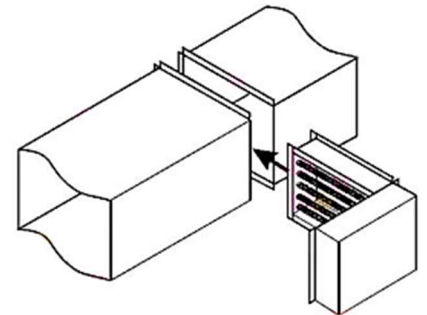
- 1- Mention heater size (WxH).
- 2- Mention Heater type (Slip-in, Flanged, Round Connection).
- 3- Mention Heater Capacity (KW), noting that our maximum heater capacity is 18 Kw.
- 4- Mention any Non-standard requirements requested by Customer. (Tubular or finned tubular, Control (2 or 3 stages or modulating), Pilot light, disconnect switch, Circuit breaker, Fuse,...).

Remarks: Note that our standard Assumptions are (Open-coil type, 1 Stage, 1 ph-220V for less than 3KW else is 3-P)

Installation:

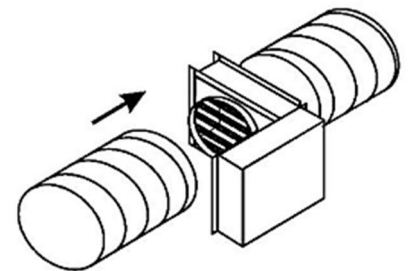
Flanged Heater (REH-R): is installed on the duct using a flange 1 inch wide on both sides

1. Position Electric heater in front of the duct flange.
2. Secure electric heater to the duct by using metal screws or bolts through the duct flanges.
3. If necessary, install supports to maintain the electric heater.



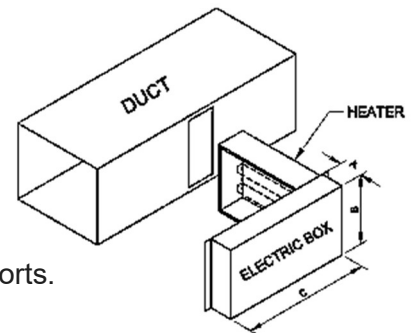
Round heater (REH-C): Hat shaped frame with a round spigot attachment, suitable for installation on circular ducts.

1. Insert electric heater between the two sections of the round duct.
2. Secure electric heater by using metal screws through round flanges.
3. If necessary, install supports to maintain the electric heater.



Slip-in heater: A side opening is made in the duct and the frame and heating coil are inserted from it by pushing them inside.

1. Cut an opening in the duct, to allow $\frac{1}{4}$ " more than the frame width.
2. Insert the heater through the opening.
3. If necessary, re-enforced the duct rigidity by installing appropriate supports.



- For more details you can scan the QR code.

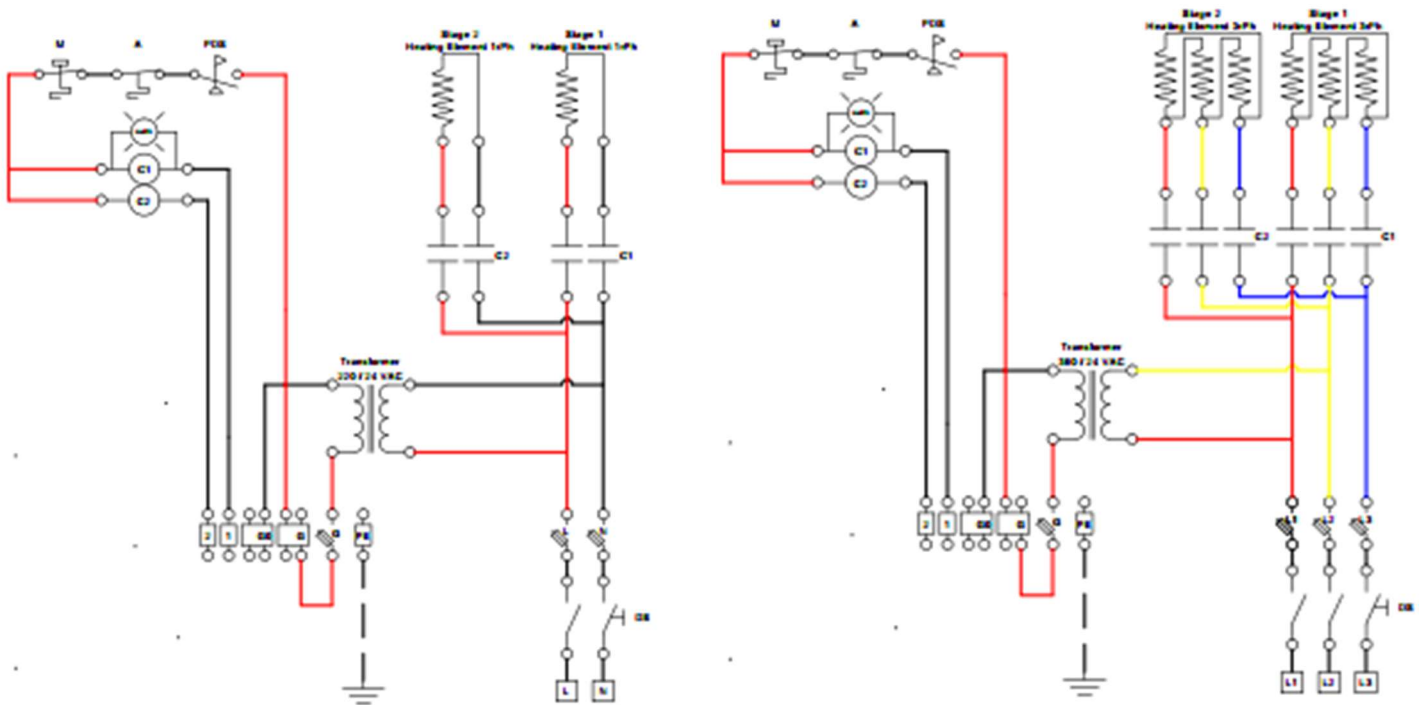


Slip-in

Electrical Installation

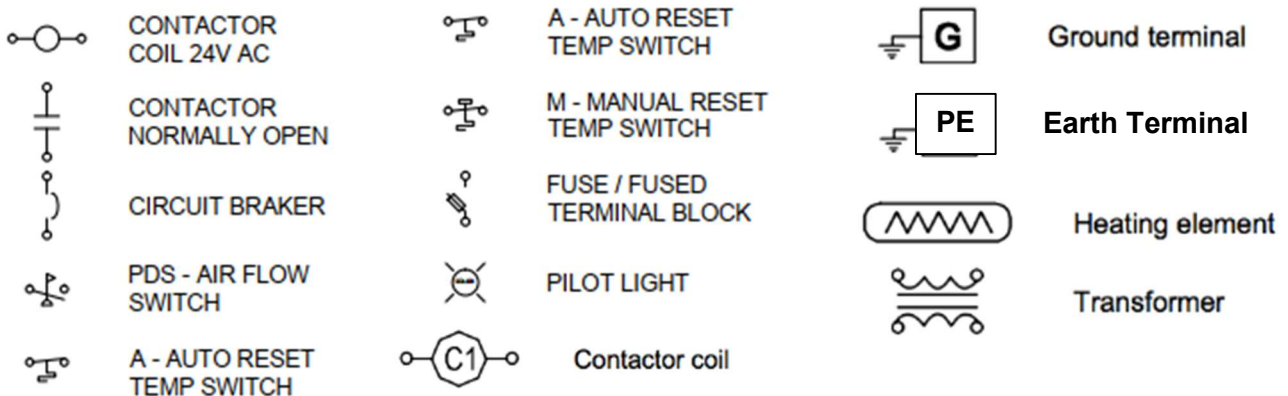
Voltage and current must match details on the heater's name plate. A wiring diagram specifically made for each heater is provided inside the control panel door. Connecting all wires to appropriate terminals as per the wiring diagram is highly important. Correct connection tightening should be verified before start up and after a short period of operation.

Wiring Diagrams



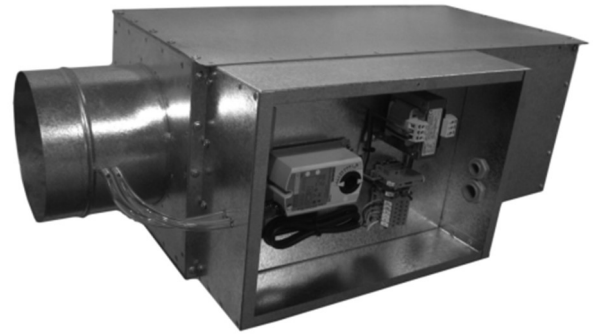
or L1 L2
L N Single phase power supply terminals

L1 L2 L3 3 phases power supply terminals



Variable Air Volume Boxes (VAV)

Variable Air Volume (VAV) boxes are an essential part of HVAC systems, and they play a critical role in regulating air distribution in different areas of a building. By adjusting the amount of conditioned air supplied to each zone, VAV boxes help to maintain appropriate temperatures and comfortable humidity levels throughout the building.



There are two main types of VAV systems:

- Single-duct VAV systems: These are the most common type of VAV systems. They use a single duct to distribute both cool and return air.
- Dual-duct VAV systems: These systems use two ducts to distribute cool and return air. This provides more flexibility for temperature control, but it is more complex and expensive than a single-duct system.
- For your reference, our product line features Single Duct VAV systems for simplified ventilation control. In addition, our VAVs are AHRI-Certified which guarantees their efficiency & reliability to our customers



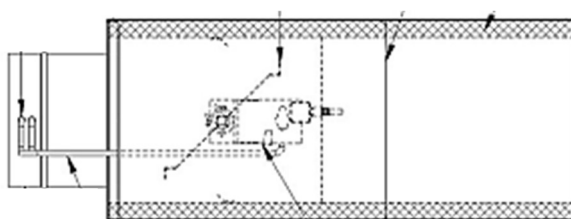
Single-duct VAV systems

There are two main categories of VAV boxes, based on how they control the amount of air that flows through them:

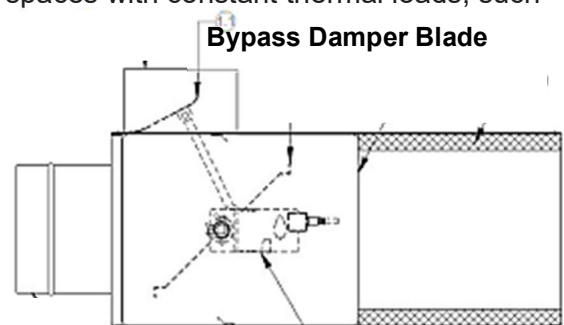
- **Bypass VAV boxes (RAV-BP):** The simplest and most common type of VAV system, typically consists of a VAV box, a bypass valve, and a thermostat. The thermostat measures the temperature in the zone and sends a signal to the bypass valve to open or close to control the amount of air supplied to each zone.
- **Pressure-independent VAV boxes (RAV-PI):** These boxes allow for more precise control of airflow. The actuator is programmed with certain airflow values, and a flow sensor measures the actual airflow through the box. The actuator then moves the damper to ensure that the actual airflow is within the programmed range.

The pressure-independent VAV boxes category is further divided into two types:

- **Supply VAV boxes (RAV-BS):** In these boxes, the actuator is programmed with two airflow values, which define the range within which the actual airflow is allowed to vary. This is the most accurate type of VAV box.
- **Constant air volume VAV boxes (RAV-BC):** These boxes are similar to supply VAV boxes, but the actuator is programmed with a single airflow value. This value is maintained throughout the operating cycle, making these boxes a good choice for spaces with constant thermal loads, such as electrical rooms.



RAV Pressure Independent Type

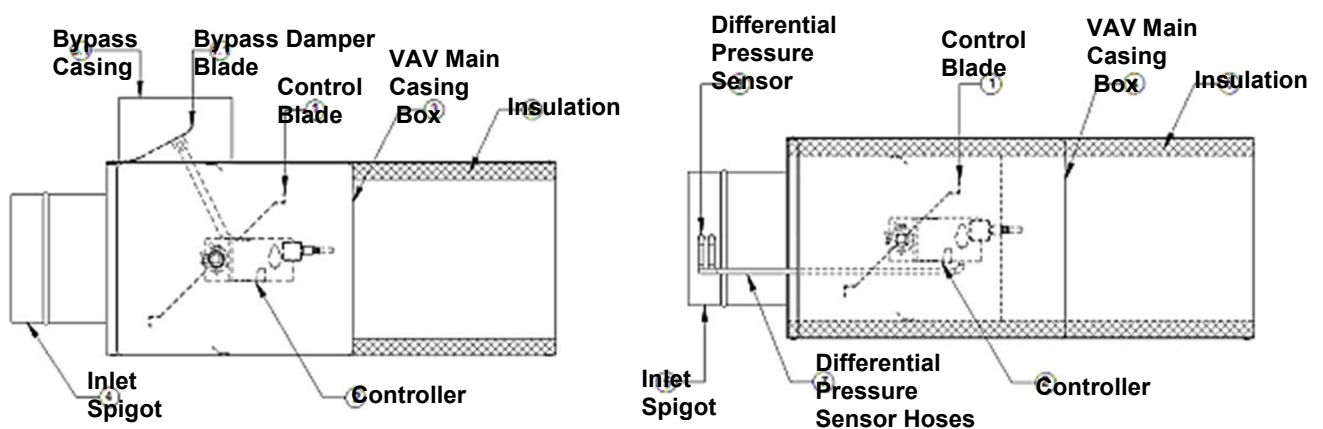


RAV Bypass Type

Construction Details:

The following components are common to all types of VAV boxes:

- **Casing:** The casing is made of galvanized steel sheet with a thickness of 1 mm. The inlet of the box is circular, and the outlet is rectangular with a flange for mounting on a standard duct.
- **Size:** The size is specified by the diameter of the inlet of the box. For sizes up to 12 inches, the inlet is circular. For sizes larger than 12 inches, the inlet is oval. The Size is indicated for each VAV box depending on the Flow rate Value.
- **Damper:** A single damper is installed inside the box, immediately after the circular inlet. The damper is the same as a standard damper in terms of specifications and installation.
- **Insulation:** The casing is lined with 1-inch thick, 24 kg/m³ density thermal and acoustic glass wool insulation. The insulation is glued in place.
- **Control box:** The control box is made of galvanized steel. It contains the electrical components necessary for controlling the circuit.



Circuit components:

- **Terminal block:** The terminal block is made of non-conductive material and has terminals for connecting the wires in the correct way.
- **Transformer:** The transformer converts the incoming electrical voltage from 220 or 380 volts to 24 volts, which is the voltage required for the control circuit.
- **Thermostat:** The thermostat is installed inside the air-conditioned space to measure the temperature inside the space. It compares the measured temperature to the desired temperature and then sends a signal to the actuator to open or close the damper accordingly. It is recommended to install the thermostat inside the space at a height of 1.5 meters from the floor.
- **Actuator:** The actuator is connected to a 24-volt power supply and receives an analog signal from the thermostat to open or close the damper according to the difference between the measured and desired temperatures.

Other options:

- **Non-standard dimensions:** The housing can be customized to non-standard dimensions upon request. However, in this case, the product is considered non-standard and the manufacturer is not responsible for any physical changes in the product's performance, such as pressure drop and noise level. However, the inlet must be circular/oval in case of pressure independent.
- **Heater:** A heater can be added to the box upon request. The specifications of the heater are the same as those of a standalone heater, with the heater size being the outlet size of the box. The heater is installed directly before the end of the box.

- **No actuator:** The product can be ordered without an actuator from the factory. In this case, the thermostat cannot be ordered unless it is confirmed to be compatible with the actuator that the customer intends to use.
- **BMS actuator:** If the box is required to be connected to a BMS or DDC Panel it has to be mentioned to use an actuator compatible with this system.
- **BMS connection:** The box can be connected to a building management system (BMS) so that it receives the signal directly from the system. This is possible if the actuator uses the same type of signal as the BMS. In this case, the thermostat can be omitted or used as a temperature indicator only.

Selection

- **Size:** The inlet size is specified based on the flow rate value as specified in the checklist, then the required size of the box is selected according to the tables shown in the catalog.

Inlet Size	Flow Rate (catalog +5%)				Outlet Size	Body length
In.	CFM	LPS	m3/s	m3/h	cm	cm
6	420	197	0.197	711	25x20	60
8	735	345	0.345	1244	25x25	70
10	1200	564	0.564	2031	35x25	70
12	1666	783	0.783	2819	45x25	70
14	2380	1119	1.119	4028	50x30	80
16	2968	1395	1.395	5022	60x30	80
18	3750	1762	1.762	6344	65x35	90
20	4711	2214	2.214	7972	80x35	90
24	6685	3142	3.142	11312	95x40	100

Model	Inlet Shape	Dx (mm)	Dy (mm)	A (mm)	F (mm)	G (mm)	L (mm)	Outlet Size (mm)	
								W	H
RAV-BP-150	Round	150	150	150	300	250	600	250	200
RAV-BP-200	Round	200	200	150	300	300	700	250	250
RAV-BP-250	Flat Oval	300	200	150	400	300	700	350	250
RAV-BP-300	Flat Oval	400	200	150	500	300	700	450	250
RAV-BP-350	Flat Oval	450	250	150	550	350	800	500	300
RAV-BP-400	Flat Oval	575	250	150	650	350	800	600	300
RAV-BP-450	Flat Oval	610	300	200	700	400	900	650	350
RAV-BP-500	Flat Oval	750	300	200	850	400	900	800	350
RAV-BP-600	Flat Oval	910	350	200	1000	450	1000	950	400

- Note that these sizes are for the VAV only, if the VAV is required to be equipped with heater the total length will be increased.

Performance

- You can determine the Detailed pressure drop values for each VAV size are readily available in the clarified table. Check Page 8 in the catalog for more details.

Model	Volumetric Flow Rate		Minimum Differential Pressure
	l/s	CFM	Pa
RAV-BS-150 RAV-BC-150 RAV-BP-150	190	400	5
RAV-BS-200 RAV-BC-200 RAV-BP-200	330	700	3
RAV-BS-250 RAV-BC-250 RAV-BP-250	540	1143	30
RAV-BS-300 RAV-BC-300 RAV-BP-300	750	1587	0
RAV-BS-350 RAV-BC-350 RAV-BP-350	1070	2267	3
RAV-BS-400 RAV-BC-400 RAV-BP-400	1335	2827	0

To find the NC value for your VAV size, select the correct table for your model in the catalog, then match your desired flow rate in the table to the corresponding size.

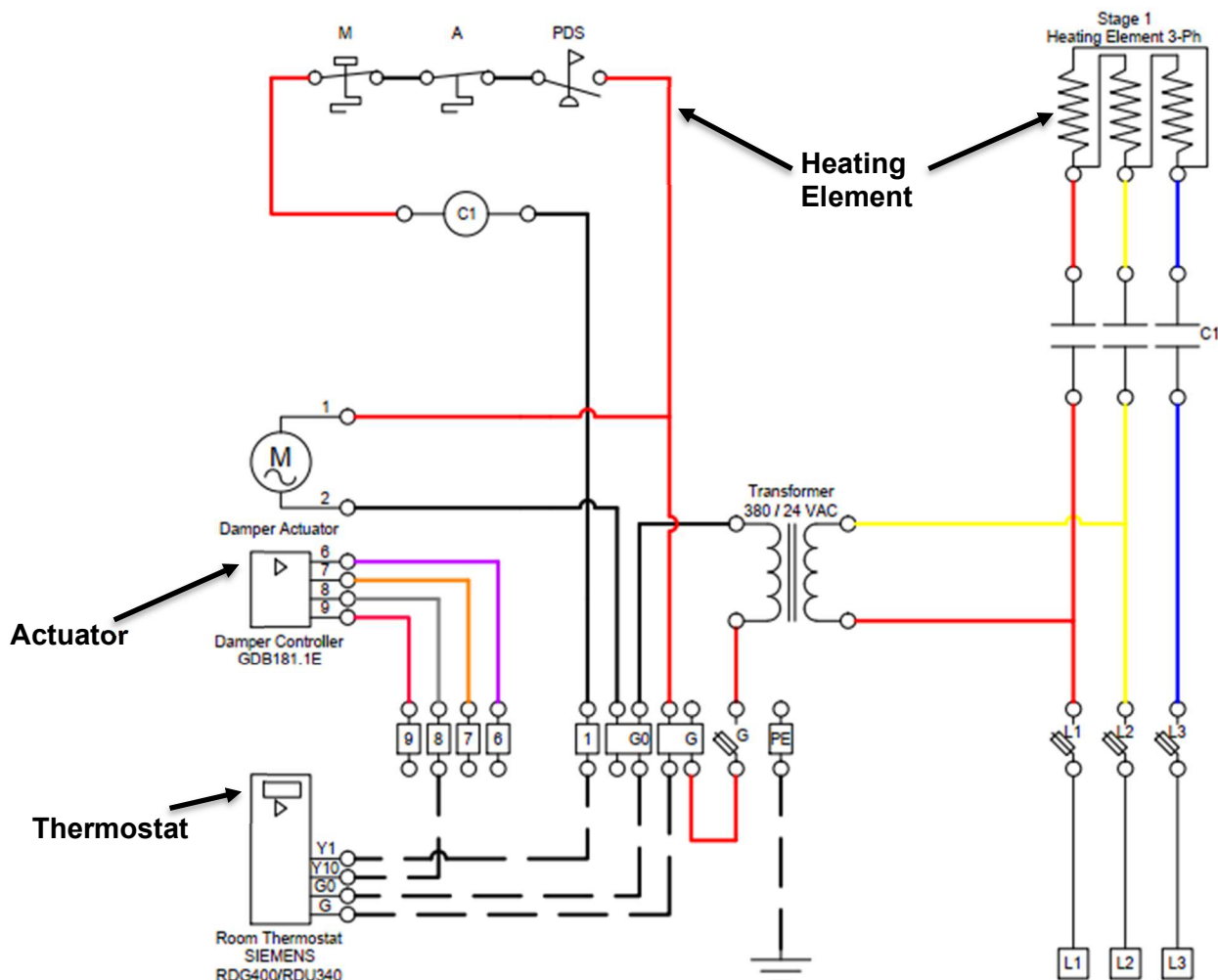
Discharge Sound Power Level

For Types **RAV-BS/RAV-BC**

Model	Airflow		Discharge Sound Power Level (dB) at Octave Band Center Frequencies (Hz) @ 125Pa P _{st}									Discharge Sound Power Level (dB) at Octave Band Center Frequencies (Hz) @ 375Pa P _{st}									Discharge Sound Power Level (dB) at Octave Band Center Frequencies (Hz) @ 500Pa P _{st}								
			LPS	CFM	63	125	250	500	1000	2000	4000	8000	NC	63	125	250	500	1000	2000	4000	8000	NC	63	125	250	500	1000	2000	4000
RAV-BS-150 RAV-BC-150	20	40	66	53	39	30	27	25	24	22	<20	68	53	45	40	35	36	42	42	<20	68	54	47	42	38	39	45	47	<20
	75	160	69	59	58	51	42	36	37	33	<20	69	64	67	63	53	46	50	50	21	69	58	60	55	50	48	52	56	<20
	130	280	72	65	63	53	45	42	40	34	<20	73	66	67	64	54	50	52	50	21	73	67	67	65	56	52	55	54	21
	190	400	75	68	66	56	49	46	43	39	22	80	74	73	68	58	54	55	54	30	78	72	72	68	60	56	57	57	27
	245	520	79	72	69	59	51	49	47	42	27	82	76	75	69	59	56	56	55	32	83	78	77	72	62	58	59	58	35
RAV-BS-200 RAV-BC-200	35	70	66	50	44	41	35	34	36	31	<20	63	50	51	46	41	41	45	44	<20	63	52	53	49	43	45	50	50	<20
	130	280	66	58	58	50	45	42	42	39	<20	66	61	61	58	51	49	52	52	<20	64	62	63	61	54	51	56	57	<20
	230	490	70	65	60	53	52	49	47	42	<20	67	70	67	61	57	55	55	55	25	68	69	68	64	58	57	58	58	23
	330	700	73	68	63	55	54	52	48	44	22	78	74	71	63	61	59	58	57	30	78	76	73	67	63	61	61	61	32
	430	910	76	71	66	58	56	53	49	45	26	81	77	73	66	63	62	60	59	34	81	79	75	69	65	64	63	62	36

Electrical Installation

A wiring diagram specifically made for each heater is provided inside the control panel door. Connecting all wires to appropriate terminals as per the wiring diagram is highly important. Correct connection tightening should be verified before start up and after a short period of operation



Differences between pressure-dependent and pressure-independent categories:

- **In pressure-dependent VAV boxes**, a bypass outlet is installed at the top of the box. A damper is installed inside the bypass outlet, and it is connected to the damper in the main airflow path by a linkage. When the main damper opens, the bypass damper closes, and vice versa. This ensures that the excess pressure in the system is relieved, thereby maintaining the system pressure.
- **In pressure-independent VAV boxes**, two aluminum tubes with an inside diameter of 8 mm are installed. Holes are evenly spaced along each tube on one side. One tube is installed so that the holes are facing the air to measure the positive pressure, and the other tube is installed so that the holes are facing away from the air to measure the negative pressure. Each tube is connected to a flexible hose that is connected to the actuator. The actuator calculates the amount of air entering the box based on the difference between the two pressure values and then responds to ensure that the actual airflow does not exceed the range specified